



Root channel macropores enhance downward movement of water in a Mojave Desert ecosystem

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Macropores formed by the root systems of woody shrubs may be an important conduit for downward water movement in desert soils. An experiment with 32 1·0-m² plots, 16 with a previous *Larrea tridentata* (creosotebush) plant and 16 without, was designed to investigate the influence of dead root systems on the downward movement of water and chemical tracers in a Mojave Desert soil. The experiment also examined the influence of water application amounts, frequency of application, and initial soil profile moisture status on downward water flux. Half of the plots received a pre-application of 15 cm of water; 48 h later either 15 cm or 30 cm of water with dye and bromide (KBr) tracers (each applied at 1·0 g l⁻¹) were applied to the plots in either a single pulse or in five equal volumes over a 5-day period. Each plot was then excavated in 10 cm depth increments 48 h after the final water and tracers were applied, and nine soil samples were taken at each depth using a 3 × 3 grid for gravimetric water and bromide contents. Gravimetric water and bromide concentration profiles with depth confirmed that water penetrated more deeply in the plots with a previous root system, and photos of the dye stained soil at each 10 cm depth confirmed that a significant percentage of the cross-sectional area was not contributing to flow at the same rate. Multiple regression analysis further indicated that the deepest depth of dye and bromide penetration, and the depth at which mean gravimetric water content was above 0·10, included the above-ground biomass of the previous *Larrea* shrub as a significant factor accounting for plot-to-plot variability. These results suggest that (1) shrub root systems act to increase downward water flux in desert soils via root channel preferential flow to deeper soil depths, and (2) this effect becomes more pronounced as shrubs and plant root systems become larger.

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Introduction

Arid climates are characterized by rates of potential evapotranspiration far exceeding precipitation on an annual basis. Even though this usually results in shallow wetting fronts in desert soils, surface moisture flux can be spatially variable due to the unique properties of desert surfaces. Many desert surfaces have well developed surface pavements and/or vesicular horizons that strongly impede the infiltration of water to functional rooting depths for perennial plants (McDonald *et al.*, 1996; Hamerlynck *et al.*,

2000). However, established woody shrubs can significantly increase infiltration of water in the vicinity of root crowns through rainfall interception, throughfall and stemflow (Lyford & Qashu, 1969; Pressland, 1976; Tromble, 1988; Mauchamp & Janeau, 1993; Martinez-Meza & Whitford, 1996). Presumably, this infiltrated water gains additional preferential flow paths in the soil (which may have relatively impermeable sub-surface layers to water infiltration; Gile *et al.*, 1998), including existing root systems and the presence of animal burrows beneath the shrub canopy. Since perennial plant cover in deserts is often 20% or less, the influence of shrub canopies (and root systems) on surface/soil moisture flux may be extremely important from a site water balance perspective.

The role of woody plants in near-surface hydrology has important applied implications in arid regions. Deserts are preferred locations for the burial of radioactive or other hazardous waste in landfills due to the lack of deep percolation of water into desert soils (Winograd, 1981; Anderson *et al.*, 1993). Soil caps for landfills containing radioactive waste are often revegetated after closure in order to minimize surface erosion and increase evapotranspiration (Waugh *et al.*, 1994; Link *et al.*, 1995). If the majority of the precipitation landing on such waste sites can be lost through the evapotranspiration process, the chances of deep percolation into zones containing the waste products would be minimized. However, the deep penetration of roots and natural die-off of plants over time could provide avenues for macropore by-pass associated with old root channels. Gish *et al.* (1998) observed that dye movement under no-tillage was more predominant in corn rows, which they felt was indicative of transport through void root channels. If these root channels provide a means to partially circumvent the soil matrix, water could arrive in a greater amount and in a shorter time than predicted by models that do not take such avenues of mass flow into account. Preferential flow associated with macropores has been demonstrated to be governed by continuity of pores (Bootlink & Bouma, 1991) and by water fluxes and pore density (Smetten, 1986; Li & Ghodrati, 1997). Kung *et al.* (2000a) demonstrated a faster arrival time and higher recovery of sequentially applied tracers, confirming that water movement and contaminant transport shifts toward increasingly larger pores of the preferential flow paths as the soil becomes wetter during a precipitation event.

Our study was conducted to determine if the intact root systems of *Larrea tridentata* (creosotebush), a Mojave Desert evergreen shrub, would enhance the downward movement of water in a desert soil system. We hypothesized that old root channels would provide by-pass (i.e. result in deeper penetration of soil water and tracers), but that such movement would be influenced by water application amounts (greater application amount resulting in greater differential penetration of water), frequency of application, and the initial soil profile moisture status.

Materials and methods

A macropore by-pass experiment was conducted in 1999 at a desert site, dominated by *L. tridentata*, near the University of Nevada's Center for Urban Water Conservation in North Las Vegas, NV (36°10'30"N; 115°08'W, 762 m). The experiment was set up to determine what influence the presence of native vegetation that had recently died (macropores associated with root channels) would have on the distance water and chemical tracers would travel in an undisturbed soil profile. Variables in the experimental design (Fig. 1) included depth of water application (15 cm or 30 cm), frequency of application (single pulse *vs.* 5 equal volume applications), and whether or not the soil began in a prewet state (15 cm of water applied 48 h before subsequent applications). Because 1999 was a drought year in the region, there were no control plots *per se* in this experiment, i.e. all 32 plots were exposed to experimental water/tracer application (Fig. 1).

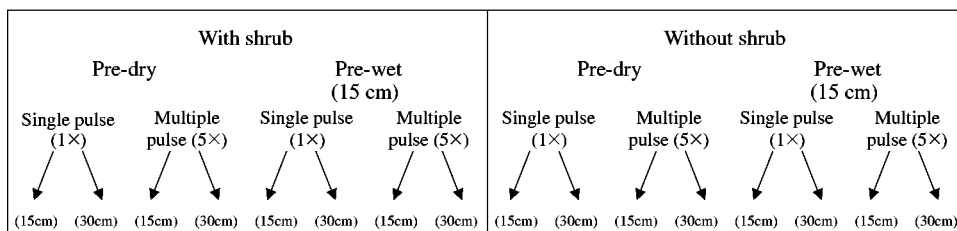


Figure 1. Experimental design of the *Larrea tridentata* preferential flow study. Two replicates per treatment.

The soil at the site was classified as a Las Vegas loam (loamy, carbonatic, thermic, shallow, typic Petrocalcic). The soil was highly calcareous with abundant fragments of calcium carbonate (greater than 2.5 cm in diameter). However, indurated layers of calcium carbonate were sporadic and generally at depths of greater than 120 cm.

Thirty-two 1.0 m² plots were selected for experimental treatment, 16 with a *L. tridentata* shrub, and 16 without. Selection of *Larrea* plots was based on identifying healthy plants of similar size and without extensive clonal development, and avoiding all plants with any observed animal burrowing activity in the near-surface soil (also for plots without vegetation). The 16 selected *Larrea* plants were then randomized for experimental treatment. Non-shrub plots were paired with each *Larrea* plot to minimize any potential surface heterogeneity across the site, but placed at least several meters away to minimize plant influences on the bare plot response. Plant height, canopy width, canopy volume, trunk diameter, stem diameter and numbers of stems were measured/counted for each of the 16 *Larrea* plants. Five months prior to initiating the experiment, all *Larrea* plants were cut at the soil surface and brought to the laboratory, where fresh and oven dry weights (70°C for 48 h) were obtained. Within 30 s of cutting the *Larrea* trunks, multiple applications of a herbicide were brushed onto the cut surfaces during a 5-min period. No re-sprouting of *Larrea* stumps was observed during the course of the experiment.

Thirty-centimeter-high soil berms were created around all of the plots. Sixteen of the plots (eight with *Larrea* and eight without) received 15 cm of water (well water, electrical conductivity 0.4 dS m⁻¹) 48 h prior to the application of dye and bromide solutions, while the other 16 plots did not receive a pre-wet application. Bromide as KBr was added to subsequent water applications at 1.0 g l⁻¹. Brilliant blue FCF dye (N-ethyl-N-[4-[[4-[ethyl[(3-sulfophenyl)methyl]-amino]phenyl](2-sulfophenyl)methylene]-2, 5-cyclohexadien-1-ylidene]-3-sulfobenzenemethanaminium hydroxide inner salt, disodium salt) was also added to subsequent water applications at 1.0 g l⁻¹. Brilliant blue dye was used to provide a visual corollary to spatial grid sampling (see below). However, the dye had a very high rate of adsorption to the calcareous loam soil, and so did not penetrate as deeply as did the wetting front and bromide tracer. Nevertheless, the dye was useful because of its distinct color contrast to the grayish white soils and because it clearly accentuated flow patterns associated with macropore water movement. After the water was gently applied to the 1 m² bermed areas (water poured into 1-l plastic containers resting on 30 cm² plastic sheeting, removed after all water was applied) the water filled plots were covered with plastic sheeting (edges of plastic secured with soil) to minimize evaporative loss. Plots receiving multiple applications of water containing dye and bromide received equal volumes of water (based on treatment) applied over a 5-day period, rather than as a single pulse.

Forty-eight hours after applying the last water to the single and multiple pulse plots, each 1 m² plot was excavated and then carefully leveled at each 10-cm depth increment with shovels and hand tools. At each 10 cm depth, including the soil surface,

photographs were taken with a digital camera and downloaded into a computer where they were transferred to a software program (Arch-View) for final analysis. Soil samples (3.87-cm auger) were then taken in a 3×3 grid with 25-cm spacing and 25 cm inside spacing from the edges of the 1 m^2 plots. Sampling continued until no soil moisture could be detected by sight or feel (no rainfall had occurred during the previous six month period, so the presence of moisture above background levels left a clear color change to the soil). Soil samples were kept cool and returned to the laboratory, where fresh and oven-dry weights were obtained. Because of the difficulty in obtaining non-compacted complete samples at all depths, soil moisture of sub-samples was calculated only on a gravimetric water content basis. A soil solution was obtained from the remaining soil using the saturation extract procedure (USSSL, 1954). Bromide was analysed in the soil solution using ion chromatography at the United States Salinity Laboratory, Riverside, CA.

The data were analysed with descriptive statistics, analysis of variance (ANOVA), and/or linear and multiple linear regression analysis. Multiple regressions were performed in a backward stepwise manner, with deletion of terms occurring when p -values for the t -test exceeded 0.05.

Results

Figure 2 shows a series of computer-enhanced photos of the dye stained cross-sectional area at multiple depths for a non-prewet (dry) single pulse 30-cm water application treatment, with and without a *Larrea* root system. As can be observed in the photos, the dye was confined to cross sectional areas of less than 100%, except for the surface, decreasing dramatically with depth and in this particular case significantly influenced by the presence of a root system. No attempt was made at quantifying the root systems because of the destructive impact it would have had on the soil sampling. However, the majority of roots at this particular site were confined to the upper 60 cm. Research by others (Wallace *et al.*, 1974) has indicated a root-to-shoot ratio for *Larrea* in the Mojave Desert of approximately 1.2. Thus plants with layer above-ground biomass could be inferred to have larger root systems.

Analysis of variance indicated an increase of 152% in the cross-sectional area stained with dye at the 10 cm depth for plots with root systems *vs.* plots without (26.7% of the surface area with roots stained with dye *vs.* 10.6% without roots; $F = 18.9$, $p = 0.0005$), and an increase in 100% in the area stained at this depth based on the amount of water applied (24.8% at 30-cm application *vs.* 12.4% at 15-cm; $F = 11.25$, $p = 0.004$). The ANOVA also indicated that the main factors influencing the depth of maximum dye penetration were the presence or absence of a root system (75% increase with roots; $F = 13.2$, $p = 0.002$), amount of water application (39% increase from 15- to 30-cm application; $F = 4.76$, $p = 0.044$), and a significant root-by-water application interaction ($F = 4.76$, $p = 0.044$). Regression analysis indicated that 43% of the variability in the deepest depth at which dye could be detected could be accounted for if the amount of water application (15 or 30 cm) and the dry weight of the aboveground plant material were included in the regression equation (max depth of dye (cm) = $4.12 + 0.87$ (water application (cm)) + 0.04 (dry weight (g)), $p = 0.001$, $n = 32$). This result would infer that larger *Larrea* plants would have a larger impact on the depth of dye movement, especially at higher application rates. In fact, for the 30-cm water application treatments with *Larrea* root systems, an excellent correlation existed between aboveground biomass and the depth of dye penetration (depth of dye penetration (cm) = $10.10 + 0.07 \times \text{aboveground biomass (g)}$, $r^2 = 0.83$).

Figure 3 shows the Br distribution with depth for the 15-cm single pulse pre-wet treatment. Both the depth of maximum Br concentration and the depth of maximum penetration were higher for the plots containing a pre-existing root system. When all 32 plots were considered, regression analysis determined that 60% of the variability in the

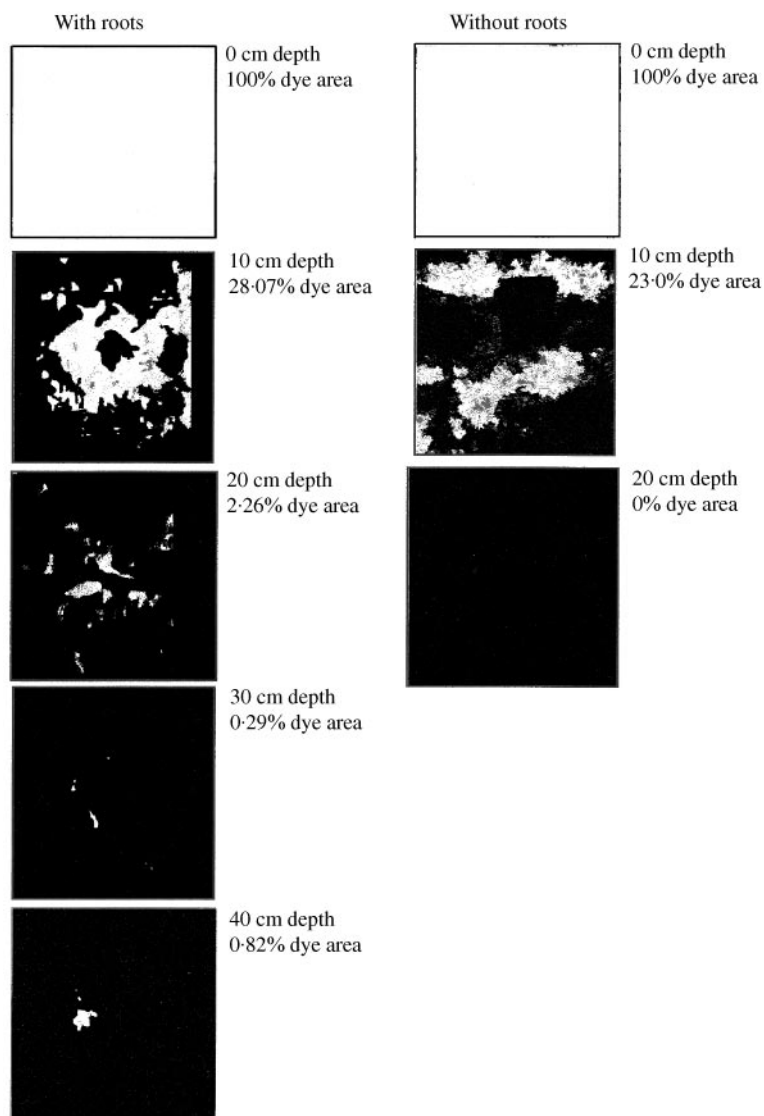


Figure 2. Percent dye stain with depth in plot 1, 30 cm single pulse pre-dry, with and without a root system.

depth at which the maximum Br concentration was detected could be accounted for if the amount of water application, frequency of application and the dry weight of the aboveground biomass were included in the regression equation (max depth of Br (cm) = $22.1 + 1.09$ (water application (cm)) - 3.74 (frequency) + 0.04 (dry weight (g)); $p = 0.001$, $n = 32$). Analysis of variance for the depth at which Br was still greater than 100 mg l^{-1} in the saturation extract, separated ($p = 0.05$ in all cases) based on amount of application (48.8 and 60.3 cm depths, respectively, at 15- and 30-cm water application), dry vs. pre-wet prior to application of tracers (49.4 and 60.0 cm depths, respectively), and absence or presence of a *Larrea* root system (47.5 and 61.9 cm depths, respectively). The bromide ion proved to be an excellent surrogate for detecting the depth of the advancing soil moisture front. Figure 4 indicates that except in one case, the

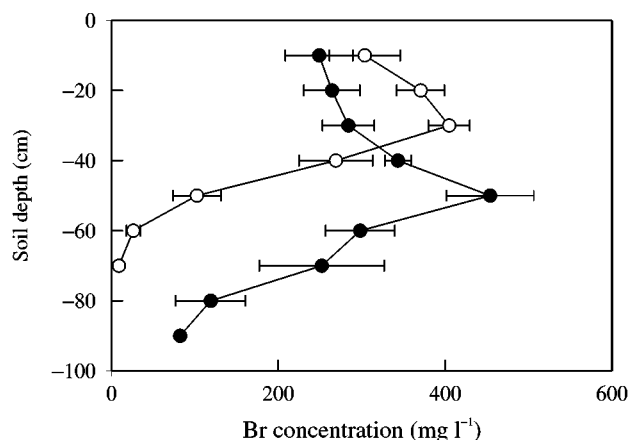


Figure 3. Average Bromide concentration with depth for the 15 cm single pulse pre-wet treatment. Horizontal bars represent the standard error of the mean ($n = 2$). With roots ($\bullet$); without roots ($\circ$).

depth at which the bromide concentration was greater than the background concentration was highly correlated (similar to the 1:1 line), with the depth at which the gravimetric water content was greater than the background level ($r = 0.81$, $n = 32$). A perfect fit was not expected since half of the plots had an initial pulse of zero bromide water applied prior to a second 15- or 30-cm pulse of water containing the bromide.

The analysis of soil moisture content with depth was the most direct test of the influence a pre-existing root system on water movement. The gravimetric water content with depth relationship for the 30-cm pre-dry treatment, with and without a *Larrea* root system, is shown in Fig. 5. Deeper penetration of the water can clearly be noted in this case when a root system was present. Multiple regression analysis for all 32 plots accounted for 68% of the variability in the maximum depth at which the gravimetric water content was greater than 0.10. The variables accepted ($p = 0.05$) in the backward

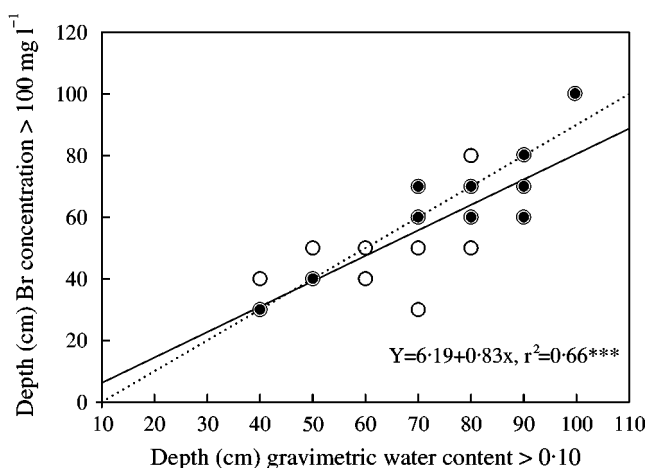


Figure 4. Maximum depth relationship between gravimetric water content (> 0.10) and bromide concentration ($> 100 \text{ mg l}^{-1}$). Solid symbols represent plots with a root system and open symbols represent plots without a root system. The solid line represents the regression relationship and the dashed line is the 1:1 relationship.

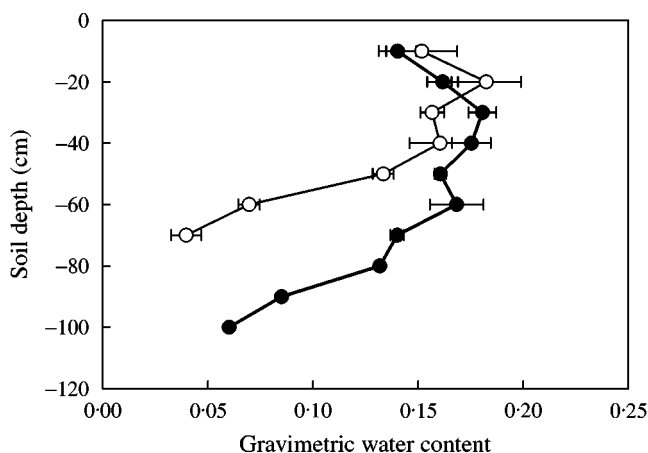


Figure 5. Gravimetric water content with depth for a 30 cm single pulse dry treatment. Horizontal bars represent the standard error of the mean ($n = 2$). With roots ($\bullet$); without roots ($\circ$).

stepwise elimination process included the amount of water application, the pre-wet status, and the dry weight of the harvested aboveground biomass (depth (cm) = $21.6 + 2.3$ (amount of water applied (cm)) + 0.045 (dry weight (g)) + 21.8 (pre-wet status; 0 = dry, 1 = wet); $p = 0.001$, $n = 32$). Analysis of variance indicated a 22% increase in the depth at which the gravimetric water content was greater than 0.10 if a root system were present (65 cm with roots *vs.* 53 cm without roots), a 28% increase in this same depth as the water application was doubled (52 and 66 cm, respectively, for the 15-cm and 30-cm application rates), and a 45% increase in this depth based on the pre-moisture status of the soil (48 and 70 cm, respectively, for the dry *vs.* pre-wet status).

Discussion

The results of this experiment provide strong evidence that the root systems of desert shrubs act as preferential flow paths for water infiltration and percolation into the soil. Our specific experimental treatments also indicated that depth of moisture and tracer penetration were generally greater when (1) the rate of water application was increased, (2) the size of the shrub (and therefore the size of the root system) was larger, (3) the soil was moist, rather than dry, prior to a large rainfall event, and (4) a given amount of water was added to the system in one large event rather than in several sequential events (over several days). Martinez-Meza & Whitford (1996) used a dye technique to trace water flow along root paths in three Chihuahuan Desert shrubs. However, they specifically examined root channel flow of stemflow water during large precipitation events (*i.e.* with intact shrub canopies). In our experiment, we removed shrub canopies to the ground surface several months prior to the application of water in order to investigate if dead or dormant root systems would channel surface water after major precipitation events. The results of the two sets of experiments clearly indicate that the root systems of desert perennials provide preferential flow paths of water to deeper soil depths.

The amount of water applied in this experiment was significantly higher than rainfall events that typically occur in southern Nevada. However, buried waste sites must be designed to safely protect the stored waste from being released to the environment over an extended period of time. This extended time (thousands of years) may be long enough to include significant climatic change. Such change could include higher total

annual rainfall, the occurrence of multiple high rainfall events over a short duration, and higher rainfall intensity runoff events that could bring significantly higher amounts of water to the waste site. The duration of the experiment was also very short in terms of moisture redistribution (4–9 days). Clearly, if higher rainfall relative to potential evapotranspiration occurred over a wider time period, deeper redistribution would be a likely consequence. However, water entry into macropores would still be closely linked with saturated flow because the wetter the soil profile, the longer the water dynamics of preferential flow paths will dominate contaminant transport (Kung *et al.*, 2000b).

Our results were based on an experimental design that attempted to minimize the variability of plant size and thus root size. However, in all cases plant size (even under these somewhat controlled conditions) became a significant factor in accounting for the variability in dye, bromide and water movement. These results would suggest that if larger plants excluded from the experiment had been included, deeper movement of moisture and tracers would have occurred. Additionally, if a longer root decay period had been allowed, we feel a more opened root channel system would have developed, leading to possibly greater macropore by-pass. It should be noted that no roots were observed in the bare plots, suggesting that the influence of near-by plants was minimized, although we cannot entirely rule out such influence without conducting a more intensive gradational study, as conducted by Dunkerly (2000), or root investigations, as conducted by Brisson & Reynolds (1994).

Mature shrubs in the Mojave Desert represent fertile islands (Schlesinger *et al.*, 1996; Whitford *et al.*, 1997) in which greater organic matter and nutrients accumulate at the base of the shrub. As such, they are also home to many insects and vertebrates that channel or burrow beneath the plant. In this experiment we eliminated plants that showed animal activity at the soil surface, even though a majority of plants revealed this association. Since these channels/burrows are continuous from the soil surface, as would be the decaying root channel system, they too would contribute to macropore by-pass under saturated flow conditions. Dunkerly (2000) observed infiltration rates 3–5 times faster near the shrub than at interspace locations associated with the Australian dryland shrub *Maireana*; this he attributed to differences in root density, textural changes, and to faunal macropores, which led to definite zones of gradational influence around shrubs not consistent with a simple binary division of zones. Gile *et al.* (1998), studying variability in root systems of *Larrea* growing in Calciargids, indicated that soil zones cemented with carbonates restricted hydraulic conductivity, funneling soil water and roots to uncemented parts of the horizon. Carbonates were present in all 32 pits excavated in our study, and although we did not quantify the extent with depth that carbonates occurred in each plot, we recognize that the extent of carbonate development could be a contributing factor to the dye, bromide and water distribution patterns observed. However, no apparent distinction could be made between plots with or without a decaying root system. Finally, Flury & Fluhler (1995) suggested that Brilliant Blue dye forms ion pairs with calcium in high calcium carbonate soils, which would support the high attenuation we observed in our soil.

In conclusion, the results from this experiment confirm that the root system of *L. tridentata* can influence the downward movement of water through the soil profile. Gish *et al.* (1998) also concluded that plant roots can be used as preferential flow pathways under till *vs.* no-till agricultural cropping systems. They observed dye distribution associated with roots that was indicative of transport through void root channels. Although the brilliant blue dye used in our experiment adsorbed quickly out of solution, it did clearly indicate flow patterns (at least in the near surface horizons) and that a significant percentage of the cross-sectional area was not contributing to flow at the same rate. This dye distribution was similar to that reported by Gish *et al.* (1998) in which both a matrix flow component, moving in an almost piston-like manner, and a preferential flow component, indicated by the fingering of dye below the matrix flow

front, existed. Interestingly, the 3×3 grid sampling within the 1 m^2 plots revealed fairly low coefficients of variation for water content (91% of sample depths with gravimetric water content, CVs less than 30%) at most depths that would not have suggested flow patterns similar to the dye results. This we believe represents the combination of slower moving fronts catching up with faster moving fronts during redistribution, especially since the dye was adsorbed out of solution so readily. However, we also believe this reveals the inadequacies of soil sampling that represents only a small percentage of the cross sectional area, as opposed to the dye assessment that was based on 100% of the cross sectional area. Such a finding does not bode well for sample number requirements in assessing characteristics of water transport, as this study was based on almost 2000 soil samples. This would be in agreement with the findings of Kung (1990), who argued that sample locations rather than the number of samples would determine how representative a group of samples is.

Finally, we conclude that if restrictive layers ('capillary breaks') are part of the cap design for aridland waste storage sites (e.g. Link *et al.*, 1995), and that rainfall characteristics do not change dramatically in the near future, the presence of vegetation would have an overwhelming positive effect on the soil-plant water balance and the minimization of a drainage component. If rainfall amounts and patterns do change in the future, care would need to be taken to minimize open channels at the soil surface that could be the result of root systems and/or animal activity.

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References

- Anderson, J.E., Nowak, R.S., Ratzlaff, T.D. & Markham, O.D. (1993). Managing soil moisture on waste burial sites in arid regions. *Journal of Environmental Quality*, **22**: 62–69.
- Bootlink, H.W. & Bouma, J. (1991). Physical and morphological characterization of bypass flow in a well structured clay soil. *Soil Science Society of America Journal*, **55**: 1249–1254.
- Brisson, J. & Reynolds, J.F. (1994). The effect of neighbors on root distribution in a creosotebush (*Larrea tridentata*) population. *Ecology*, **75**: 1693–1702.
- Dunkerly, D. (2000). Hydrologic effects of dryland shrubs: defining the spatial extent of modified soil water uptake rates at an Australian desert site. *Journal of Arid Environments*, **45**: 159–172.
- Flury, M. & Fuhler, H. (1995). Tracer characteristics of Brilliant Blue FCF. *Soil Science Society of America Journal*, **59**: 22–27.
- Gile, L.H., Gibbens, R.P. & Lenz, J.M. (1998). Soil-induced variability in root systems of creosotebush (*Larrea tridentata*) and tarbush (*Flourensia cernua*). *Journal of Arid Environments*, **39**: 57–78.
- Gish, T.J., Gimenez, D. & Rawls, W.J. (1998). Impact of roots on ground water quality. *Plant & Soil*, **200**: 47–54.
- Hamerlynck, E.P., McAuliffe, J.R. & Smith, S.D. (2000). Effects of surface and sub-surface soil horizons on the seasonal performance of *Larrea tridentata* (creosotebush). *Functional Ecology*, **14**: 596–606.
- Kung, K-J.S. (1990). Preferential flow in a sandy vadose zone: 1. Field observation & 2. Mechanism and implications. *Geoderma*, **46**: 51–71.
- Kung, K-J.S., Kaladivko, E.J., Gish, T.J., Steenhuis, T.S., Bubenzer, G. & Helling, C.S. (2000a). Quantifying preferential flow by breakthrough of sequentially applied tracers: Silt loam soil. *Soil Science Society of America Journal*, **64**: 1296–1304.
- Kung, K-J. S., Steenhuis, T.S., Kaladivko, E.J., Gish, T.J., Bubenzer, G. & Helling, C.S. (2000b). Impact of preferential flow on the transport of adsorbed and non-adsorbing tracers. *Soil Science Society of America Journal*, **64**: 1290–1296.

- Li, Y. & Ghodrati, M. (1997). Preferential transport of solute through soil columns containing constructed macropores. *Soil Science Society of America Journal*, **61**: 1308–1317.
- Link, S.O., Wing, N.R. & Gee, G.W. (1995). The development of permanent isolation barriers for buried wastes in cool deserts: Hanford, Washington. *Journal of Arid Land Studies*, **4**: 215–224.
- Lyford, F.P. & Qashu, H.S. (1969). Infiltration rates as affected by desert vegetation. *Water Resources Research*, **5**: 1373–1376.
- McDonald, E.V., Pierson, F.B., Flerchinger, G.N. & McFadden, L.D. (1996). Application of a process based soil-water balance model to evaluate the influence of Late Quaternary climate change on soil-water movement. *Geoderma*, **74**: 167–192.
- Martinez-Meza, E. & Whitford, W.G. (1996). Stemflow, throughfall and channelization of stemflow by roots in three Chihuahuan Desert shrubs. *Journal of Arid Environments*, **32**: 271–287.
- Mauchamp, A. & Janeau, J.L. (1993). Water funneling by the crown of *Flourensia cernua*, a Chihuahuan Desert shrub. *Journal of Arid Environments*, **25**: 299–306.
- Pressland, A.J. (1976). Soil moisture redistribution as affected by throughfall and stemflow in an arid zone shrub community. *Australian Journal of Botany*, **24**: 641–649.
- Schlesinger, W.H., Raikes, J.A., Hartley, A.E. & Cross, A.F. (1996). On the spatial pattern of soil nutrients in desert ecosystems. *Ecology*, **77**: 364–374.
- Smetten, K.R.J. (1985). Analysis of water flow from cylindrical macropores. *Soil Science Society of America Journal*, **50**: 1139–1142.
- Tromble, J.M. (1988). Water interception by two arid land shrubs. *Journal of Arid Environments*, **15**: 65–70.
- USSL (U.S. Salinity Laboratory Staff.) (1954). *Diagnosis and Improvement of Saline and Alkali Soils*. USDA Agricultural Handbook 60. U.S. Government Printing Office, Washington, D.C., USA.
- Wallace, A., Bamberg, S.A. & Chow, J.W. (1974). Quantitative studies of roots of perennial plants in the Mojave Desert. *Ecology*, **55**: 1160–1162.
- Waugh, W.J., Thiede, M.E., Bates, D.J., Caldwell, L.L., Gee, G.W. & Kemp, C.J. (1994). Plant cover and water balance in gravel admixtures at an arid waste-burial site. *Journal of Environmental Quality*, **23**: 676–685.
- Whitford, W.G., Anderson, J. & Rice, P.M. (1997). Stemflow contribution to the 'fertile island' effect in creosotebush, *Larrea tridentata*. *Journal of Arid Environments*, **35**: 451–457.
- Winograd, I.J. (1981). Radioactive waste disposal in thick unsaturated zones. *Science*, **212**: 1457–1464.